# Shannon McClish

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## About

I am a biogeochemical oceanographer interested in using observations and models to understand the physical processes that influence ocean carbon, oxygen, and nutrient cycling in a changing world. I recently finished my PhD at the University of Hawai'i at Mānoa where I investigated biological carbon cycling with biogeochemical Argo float and satellite observations. I will soon be starting as a postdoctoral researcher at the University of British Columbia where I will be using glider observations to investigate how turbulence modulates oxygen concentrations in the NE Pacific.

## Education

**PhD University of Hawai'i at Mānoa**, Oceanography. — Aug 2019 – Sept 2025

- Dissertation: *Resolving seasonal to annual marine net community production using biogeochemical argo float observations.* Supervisor: Professor Seth Bushinsky

**BS University of California, Santa Cruz** Earth Science with highest honors. — Aug 2015 – June 2019

- Bachelors Thesis: *Role of Circulation Changes on the Carbon Cycle Anomaly during Eocene Thermal Maximum 2: Insights from Observations and Modeling.* Supervisors: Professor James Zachos, Professor Mathis P. Hain

**ETH Zurich** Visiting Student-Department of Earth and Planetary Science. — Aug 2017 – June 2018

- Research Project: *Assessing the use of Uk37 and TEX86 Sea Surface Temperature Proxies in Upwelling systems.* Supervisor: Dr. Blanca Ausín González

## Experience

**C-PROOF and PRODIGY**, Postdoctoral Research Fellow. — Vancouver, BC, Oct 2025
Supervisor: Professor Stephanie Waterman, swaterman@eoas.ubc.ca

- Leverage glider observations to study local drivers of deoxygenation in coastal British Columbia as part of the Canadian-Pacific Robotic Ocean Observing Facility (C-PROOF) program.
- Train and mentor graduate students in ocean observation and prediction as part of Pacific Rim Ocean Data Mobilization and Technology (PRODIGY) program.
- Lead field deployments of turbulence-sensing gliders in coastal British Columbia.

**HI-CYCLES lab**, Graduate Research Assistant. — Honolulu, HI, Jan 2020 – Sept 2025
Supervisor: Professor Seth Bushinsky, seth.bushinsky@hawaii.edu

- Co-developed outlier detection and secondary quality control procedures of biogeochemical Argo float data. https://www.bgc-argo-plus.info/
- Mentored an undergraduate student for their undergraduate senior thesis, and assisted in securing funding for their work.
- Co-created outreach activities and engaged in lab community outreach.

**GO-SHIP A16N**, CTD Operations — Apr 2022 – May 2022

- Assisted with CTD operations and sampling on the 2022 GO-SHIP A16N leg 2 from Spain to Iceland.

**Schmidt Ocean Institute**, Science Communication Intern — Honolulu, HI, Fall 2019 – Fall 2020
Supervisor: Dr. Carlie Wiener, cwiener@schmidtocean.org

- Authored articles, magazine pieces, and press releases, assisted with community outreach events, managed social media accounts, and facilitated ship to shore

classroom connections.

**Department of Oceanography**, Teaching Assistant — Honolulu, HI
Supervisor: Professor Margaret McManus mamc@hawaii.edu — Aug 2019 – May 2020

- Taught an introductory "Science of the Sea" lab that included a small lecture component, lab activities, and a field trip covering topics in physical, biological, and chemical oceanography.
- Assisted in teaching an upper-division undergraduate biogeochemistry course with Matlab box-modeling and problem sets.

**Zachos lab**, Undergraduate Lab Assistant. — Santa Cruz, CA
Supervisor: Professor James Zachos — Fall 2018 – Summer 2019

- Identified select foraminifera species, prepared various samples and conducted stable isotope analysis.

**ETH Zurich Climate Geology Lab**, Undergraduate Lab Assistant. — Zurich, CH
Supervisor: Professor Heather Stoll — Winter 2018 – Summer 2018

- Assisted post-doctoral research with laboratory duties preparing various types of geologic samples for stable isotope analysis.

## Publications

McClish, S., Bushinsky, S. M., Briggs, N., Douglas, C. (submitted). Seasonal re-entrainment of respired organic matter decouples surface production and annual net community production in the Southern Ocean. preprint here

McClish, S., & Bushinsky, S. M. (2023). Majority of Southern Ocean seasonal sea ice zone bloom net community production precedes total ice retreat. Geophysical Research Letters, 50(20), e2023GL103459. access here

Weiner, C., & McClish, S. (2019). Exploring New Worlds Here on Earth. Eco Magazine, Nov/Dec 2019, 26–29. access here

Wiener, C., McClish, S., Mock-Bunting, L, and Virmani, J. (2020). Exploration and Innovation at Sea: The Schmidt Ocean Institute 2019 Field Season. Oceanography 33(1), supplement, 102-109. access here

## Fellowships and awards

**NASA Future Investigator in Earth System Science and Technology award** — 2021-2024

24NNH20ZDA001N-FINESST- Characterizing the impact of seasonal sea ice on phytoplankton blooms and net community production with integrated satellite and biogeochemical profiling float observations.

**ARCS foundation George and Mona Elmore Award in Oceanography** — 2022
**University of California Santa Cruz Holly Day Barnett Fund** — 2019
**University of California Regents Scholar** — 2015-2019

## Service

- University of Hawai'i Graduate Student Organization Sustainability Chair *2021-2024*
- Co-creator and mentor for the Department of Oceanography graduate student admissions mentorship program *2023-2024*
- UH Mānoa Undergraduate research showcase volunteer and moderator *Fall 2023, Spring 2024*
- Department of Oceanography Na Kama Kai Officer *2019-2024*
- University of Hawai'i Graduate Student Organization Representative *2019-2021*

## Outreach

- Planned science communication workshop for PRODIGY graduate students from Canada and Chile *2025*
- Served on a graduate student panel for science of the sea undergraduate course (once a semester) *2023-2025*
- Outreach and mentor for local high school students with Graduate women in science at UH Mānoa (once a semester) *2024*
- Lego profiling float outreach activity at UH Discover Mānoa event *Fall 2024*
- Central Coast State Parks Association Mind walk lecture on oceanography of the central coast of California, virtual public lecture *Spring 2024*
- Lego profiling float SOEST Open house outreach activity with local community *Spring 2023*
- Outreach with elementary school students in Watsonville CA (virtual) *Spring 2022, Fall 2020*
- Weekly in person and virtual outreach with K-12 students and general public while working as the Schmidt Ocean Institute Science Communications Intern. Activities ranged from assisting with research vessel tours, ROV livestreams, Classroom to ship connections, and writing blog and social media posts. *Fall 2019- Fall 2020*

## Selected Presentations

**NASA Ocean Biology and Biogeochemistry All hands meeting** Oral Presentation-Seasonal re-entrainment of respired organic matter decouples surface and annual net community production in the Southern Ocean 2024

**Southern Ocean Carbon and Climate Observations and Modeling Project Annual Meeting** Oral Presentation-Dynamics of Net Community Production in the Southern Ocean 2024

**Ocean Sciences Meeting** Oral Presentation-Dynamics of Net Community Production in the Southern Ocean 2024

**Southern Ocean Carbon and Climate Observations and Modeling Project Annual Meeting** Oral Presentation-Investigating net community production in the Southern Ocean seasonal sea ice zone. 2023

**Southern Ocean Observing System Symposium** Poster-Majority of spring bloom net community production in Southern Ocean seasonal sea ice zone occurs before sea ice retreat 2023

**Ocean Carbon and Biogeochemistry Summer Workshop** Poster-Characterizing the impact of seasonal sea ice on phytoplankton blooms and NCP in the Southern Ocean. 2022

**Southern Ocean Carbon and Climate Observations and Modeling Project Annual Meeting** Oral presentation-Characterizing the impact of seasonal sea ice on phytoplankton blooms and NCP in the Southern Ocean. 2022

**NASA Ocean Color Research virtual meeting** Oral Presentation (remote)-Characterizing the impact of seasonal sea ice on phytoplankton blooms and upper ocean biogeochemistry with integrated satellite and profiling float observations 2021